

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)		Constant velocity (1) Substitution of $D = vT$ into Hubble's law and convincing algebra (1)	2			2	1	
	(b)		Substitution of H_0 & 3.09×10^{22} m into Hubble's law (gives 68 000 m/s) (1) Substitution [of at least c and λ] into Doppler equation (1) $\Delta\lambda = \frac{68\,000 \times 486.1 \times 10^{-9}}{3 \times 10^8}$ seen (1) or 0.11 nm	1 1	1		3	3	
	(c)		Correct method for obtaining velocities e.g. $\frac{c\Delta\lambda}{\lambda}$ (1) Blue shift speed is $[-]2v$ (-136 000) (1) and redshift speed is $6v$ (407 000 / 408 000) (1) Recessional speed = $2v$ (136 000) (1) Rotational speed = $4v$ (272 000) (1) NB Final mark awarded only if recessional speed and rotational speed clearly identified.		5		5	5	
	(d)		Collation points C1 Doppler/red/blue shift measured C2 of [known] lines / wavelengths C3 velocity calculated / measured / linked to $\Delta\lambda / \frac{\Delta\lambda}{\lambda} = \frac{v}{c}$ C4 for various distances from centre of galaxy [can be credited from good graph] C5 absorption spectrum used C6 mention of [large] telescope / analysis of light / em radiation observed C7 mention of spectrometer / prism / grating / spectral analysis Theory points T1 Doppler shift explained e.g. redshift, moving away T2 Doppler equation quoted or shift dependency on velocity mentioned T3 Orbital speed increases with mass / $v = \sqrt{\frac{GM}{r}}$ T4 Orbital speed decreases with radius (theoretically) OR graph	6			6		